Failure of Bayesian Gaussian process classification at exponentially large sample sizes

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Abstract

We study the exact Gaussian sign posterior for the saturated neural-network Gaussian process on the Boolean cube. At every fixed depth, we prove that there is an exponentially large sample schedule on which learning full parity still leaves a fixed positive fraction of inputs without correct-label probability greater than 2/3, with probability at least 3/4 over independent uniform training inputs. The result also applies to the width limit of globally margin-conditioned Bayesian tanh networks at every fixed scaling exponent $\gamma < 0$. The proof combines sharp Fourier level masses, a random tail-covariance operator bound, and uniform ReLU sampling with exact constraints on the posterior mean. It concerns the posterior predictive probability, rather than the sign of a regression mean. The conclusion is failure at specified exponential sample sizes; a lower bound covering every smaller sample size is not proved. The fixed-margin case $\gamma = 0$ and the regime $0 < \gamma < 1$ are outside the exponential result established here.

1 Model, result, and scope

Let $\mathcal{X} = \{-1, 1\}^d$ with the uniform probability measure, and let $f(x) = \prod_{j=1}^d x_j$ be full parity. Although parity has Boolean circuits of size $O(d)$, its learnability depends on the specified prior and prediction rule. We analyze Gaussian-process classification with covariance

$$K(x, y) = H^{\circ(L-1)}\left(\frac{1 + x \cdot y}{d + 1}\right), \qquad H(t) = \frac{2}{\pi} \arcsin t, \tag{1}$$

where the depth $L \geq 2$ is fixed. For a centered Gaussian vector $G \sim N(0, K)$ and training inputs $X_1, \ldots, X_n$, the posterior conditions on the exact signs $f(X_i)G(X_i) > 0$. Its probability of being correct at x is

$$q_S(x) = \mathbb{P}\{f(x)G(x) > 0 \mid f(X_i)G(X_i) > 0 \text{ for all } i\}.$$

We count an input as a failure when $q_S(x) \leq 2/3$. Thus the criterion concerns the actual posterior probability and includes equality as a failure. It is not identified with the error of the sign of a posterior mean.

Our main theorem, Theorem 6.1, gives constants $A, \alpha, \varepsilon > 0$ depending at most on the fixed depth. With r_d odd and $r_d/d \to \alpha$, put

$$D_{r_d} = \sum_{j=0}^{r_d} \binom{d}{j}, \qquad n_d = \lceil A D_{r_d} \log(D_{r_d} + 1) \rceil. \tag{2}$$

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Then $n_d = \exp((h(\alpha) + o(1))d)$, where h is the binary entropy with natural logarithms. With probability at least $3/4$ over n_d independent uniform training inputs, a fraction at least ε of test inputs has correct-label probability at most $13/20 < 2/3$. One conservative choice in the proof is

$$\varepsilon = \frac{2\pi(1/e - 7/20)^2}{2048} > 9 \cdot 10^{-7}.$$

The constants are not optimized. In particular the theorem establishes a fixed positive failure fraction, not a numerical failure fraction such as $1/10$.

The quantifiers in this conclusion matter. It establishes failure on the displayed exponential sample schedule and does not assert failure at every smaller sample size. Posterior confidence for a fixed target has not been shown to be monotone in the sample size. The argument also does not establish an exponential statement at zero or positive subcritical scaling exponents. The later paper [1] proves failure at every specified sample size below an exponential threshold, and [2] also treats zero and positive subcritical exponents. This paper retains the earlier schedule argument and its quantifiers.

1.1 The finite-width network and its Gaussian limit

The same theorem applies to the following network convention. All hidden layers have width N, with

$$h^0(x) = (1, x), \quad h^s(x) = \tanh(W^s h^{s-1}(x)) \quad (1 \le s < L), \quad z_N(x) = W^L h^{L-1}(x).$$

There are no additional biases. The weights are independent centered Gaussians with variances

$$\mathrm{Var}(W^1_{ij}) = N^{1-\gamma}/d, \quad \mathrm{Var}(W^s_{ij}) = N^{-\gamma} \ (1 < s < L), \quad \mathrm{Var}(W^L_i) = N^{-\gamma-1}. \tag{3}$$

Equivalently, prescribe the normalized weight scale $t(N) = N^{(1-\gamma)/2}$ before varying width. The three variances are $t(N)^2/d$, $t(N)^2/N$, and $t(N)^2/N^2$, respectively. Thus γ is fixed while the scalar scale follows this width schedule.

Before training, condition on the global margin event $A_a = \{|z_N(x)| > a \text{ for every } x \in \mathcal{X}\}$, where $a > 0$ is fixed. Training then conditions on $\operatorname{sign} z_N(X_i) = f(X_i)$. Let $q_{N,S}(x)$ be the resulting correct-label probability and $F_N(S) = \mathbb{P}_x\{q_{N,S}(x) \le 2/3\}$.

Proposition 1.1 (Conditional width limit)**.** *At fixed d, L, a, S and $\gamma < 0$, for every $x \in \mathcal{X}$,*

$$q_{N,S}(x) \longrightarrow q_S(x).$$

All these posteriors are well defined for sufficiently large N. Consequently

$$\liminf_{N\to\infty} F_N(S) \ge \mathbb{P}_x\{q_S(x) \le 13/20\}.$$

Proof. Let $Q^s_N(x, y) = N^{-1}\sum_i h^s_i(x)h^s_i(y)$. Conditional on the preceding layer, hidden rows are independent bounded random vectors. The conditional law of large numbers therefore applies to all empirical second moments, since the cube is finite. Preactivation variances grow like $N^{1-\gamma}$. Thus tanh converges to the sign function under a Gaussian coupling. The identity $\mathbb{E}[\operatorname{sign} U \operatorname{sign} V] = (2/\pi)\arcsin\rho$ for standard jointly Gaussian U, V of correlation ρ follows, for example, by measuring the corresponding angular sectors of a rotationally invariant planar Gaussian. Dominated convergence and induction over layers yield

$$Q^{L-1}_N \longrightarrow K \quad \text{in probability.}$$

For completeness, the convergence of the conditional moment map is uniform near its limiting covariance: if it were not, choose a converging covariance subsequence, couple by continuous matrix square roots, and apply dominated convergence. Its diagonal is positive, so the limiting scalar Gaussian coordinates vanish with probability zero. The first layer normalizes $(d+1)/d$ on the diagonal and hence gives the argument $(1+x\cdot y)/(d+1)$ in (1).

Conditionally on the hidden layers, $N^{\gamma/2}z_N \sim N(0, Q_N^{L-1})$. Therefore the whole output vector $G_N = N^{\gamma/2}z_N$ converges in law to $G \sim N(0,K)$. The global margin in these coordinates is $|G_N(x)| > aN^{\gamma/2}$, whose threshold tends to zero. The spectral proof below shows K is positive definite. Hence all coordinate hyperplanes are Gaussian null sets and every sign orthant has positive probability. A shrinking-neighborhood argument around the coordinate hyperplanes shows that imposing this vanishing margin changes the probability of each fixed sign event by $o(1)$. Ratios of probabilities of sign events now give $q_{N,S}(x) \to q_S(x)$ and positivity of the conditioning probability for large N.

At a point where $q_S(x) \le 13/20$, convergence implies $q_{N,S}(x) < 2/3$ eventually. The cube is finite, so the implication holds simultaneously at all these points, proving the final assertion. □

The width limit is always taken at fixed d, L, γ, a, S. The subsequent exponential statement concerns $d \to \infty$ for these limiting learners. No uniform interchange of the width and dimension limits is asserted.

Proof roadmap. The Fourier estimates control both the trace and largest eigenvalue of the high-degree covariance. Random-submatrix bounds then control that covariance on the sample. Convex Gaussian sign conditioning bounds the actual posterior mean and turns confident predictions into latent margins. Uniform sampling of its low-degree part completes the contradiction at the specified sample schedule.

2 Fourier level masses of the kernel

Put $D = d+1$, $\beta = 2^{-(L-1)}$, and $F = H^{\circ(L-1)}$. The characters $\chi_U(x) = \prod_{j\in U} x_j$ diagonalize the normalized covariance operator $M^{-1}K$, $M = 2^d$. Let λ_j be its eigenvalue at degree j. We first work on the full D-cube with kernel $F(u\cdot v/D)$, whose degree-j eigenvalue is denoted by a_j. Constants in this and the next section may depend on L. Notation $b \asymp c$ means two inequalities by positive constant multiples, uniformly in the stated ranges.

Lemma 2.1 (Power coefficients and paired eigenvalues)**.** *Write $F(z) = \sum_{m \text{ odd}} c_m z^m$. Every odd coefficient is positive, $\sum_m c_m = 1$, and*

$$c_m \asymp \frac{\Gamma(m-\beta)}{m!} \asymp m^{-1-\beta} \qquad (m \ge 1 \text{ odd}).$$

The full-cube eigenvalues vanish at even degrees and strictly decrease through odd degrees. They are positive at every odd degree. On the original d-cube,

$$\lambda_{2k} = \lambda_{2k+1} = a_{2k+1} > 0,$$

with a final unpaired eigenvalue if d is even. In particular K is positive definite.

Proof. The series of H has positive coefficients at every odd exponent, with

$$[z^{2k+1}]H(z) = \frac{2}{\pi}\frac{\binom{2k}{k}}{4^k(2k+1)}.$$

Composition preserves strict positivity and oddness. Near $z = 1$ in a slit neighborhood,

$$H(z) = 1 - b(1-z)^{1/2} + O((1-z)^{3/2}), \qquad b = \frac{2\sqrt{2}}{\pi}.$$

Induction gives $F(z) = 1 - A(1-z)^{\beta} + O((1-z)^{3\beta})$, with $A > 0$; at each composition its leading constant changes from A_t to $b\sqrt{A_t}$, starting from $A_1 = b$. These expansions admit algebraic coefficient transfer [3, Chapter VI]. Here are the analytic-domain details. On the unit circle outside $\{-1, 1\}$, positivity of every odd coefficient gives $|H(z)| < 1$: equality in the triangle inequality would require $z^2 = 1$. Hence the iterated functions extend analytically through every compact arc away from these endpoints. At 1, restrict $\arg(1-z)$ to a closed proper subinterval of $(-\pi, \pi)$. Each square-root composition reduces this angular range, and its positive leading coefficient keeps its image in the slit neighborhood for the next arcsine composition. Oddness gives the identical argument at -1. These local sectors and the compact arcs provide the required two-singularity indented domain. The contributions of 1 and -1 add at odd exponents, giving

$$c_m \sim \frac{2A}{-\Gamma(-\beta)} m^{-1-\beta} \quad (m \to \infty \text{ odd}).$$

Together with positivity of the finitely many initial coefficients and the gamma-ratio asymptotic, this proves the coefficient comparisons. For $L = 2$ the explicit formula and Stirling's bounds suffice instead.

Let $p_m(j)$ be the probability that the simple coordinate-flip walk on the D-cube reaches a specified distance-j vertex in m steps. Expanding $(u \cdot v/D)^m$ shows that its degree-j Fourier coefficient is $p_m(j)$. Thus $a_j = \sum_{m \text{ odd}} c_m p_m(j)$. For the walk recursion,

$$\begin{aligned} D[p_{m+1}(j) - p_{m+1}(j+2)] = {} & j[p_m(j-1) - p_m(j+1)] \\ & + (D-j-2)[p_m(j+1) - p_m(j+3)]. \end{aligned}$$

Induction from time zero gives nonincreasing probabilities through each parity class; terms with zero boundary coefficients are omitted. The term $m = j$ makes the odd-eigenvalue inequalities strict and proves positivity. Even-degree eigenvalues vanish by parity. Restricting the distinguished coordinate to $+1$ gives $\lambda_q = a_q + a_{q+1}$, proving the claims. □

For odd r, the exact combined multiplicity of reduced degrees $j-1, j$ is $\binom{d}{j-1} + \binom{d}{j} = \binom{D}{j}$. Consequently

$$D_r = \sum_{\substack{j \le r \\ j \text{ odd}}} \binom{D}{j}, \quad \tau_r = \sum_{\substack{j > r \\ j \text{ odd}}} \binom{D}{j} a_j, \quad \lambda_{r+1} = a_{r+2}. \tag{4}$$

The odd cutoff is essential for these exact identities.

Lemma 2.2 (Sharp level and tail estimates)**.** *Uniformly for odd* $1 \le j \le D/4$ *and odd* $1 \le r \le D/8$,

$$\binom{D}{j} a_j \asymp j^{-1-\beta}, \tag{5}$$

$$\tau_r \asymp r^{-\beta}, \tag{6}$$

$$\frac{\lambda_{r+1}}{\tau_r} \asymp \frac{1}{r\binom{D}{r+2}}, \tag{7}$$

$$s_{2,r} := \sum_{q > r} \binom{d}{q} \lambda_q^2 \le \frac{C r^{-2\beta-2}}{\binom{D}{r+2}}. \tag{8}$$

The same level-mass estimate holds up to $j \le \rho D$ for any fixed $\rho < 1/2$, with constants depending on ρ.

Proof. The gamma integral and Lemma 2.1 give, for odd j,

$$a_j \asymp D^{-\beta} \int_0^\infty u^{-1-\beta} p(u)^j (1-p(u))^{D-j}\, du, \quad p(u) = \frac{1-e^{-2u}}{2}, \tag{9}$$

$$\asymp D^{-\beta} \int_0^{1/2} p^j (1-p)^{D-j} w_\beta(p)\, dp, \tag{10}$$

where $w_\beta(p) = [-\frac{1}{2}\log(1-2p)]^{-1-\beta}/(1-2p)$. Indeed the gamma integral Poissonizes the walk time, and independent Poisson coordinate counts give independent coordinate parities with success probability $p(u)$. Since an odd-distance vertex has zero probability at even walk times, no extra parity factor occurs. The weight w_β is integrable near $1/2$ and comparable to $p^{-1-\beta}$ on $[0, b]$ for every fixed $b < 1/2$.

Fix $\rho < b < 1/2$ and take $j \le \rho D$. The integral on $[0, b]$ is comparable to the complete beta integral $B(j-\beta, D-j+1)$. For the lower comparison, a beta random variable with these parameters has mean at most ρ, so Markov's inequality puts at least a fraction $1 - \rho/b$ of its mass in $[0, b]$. The identity

$$\binom{D}{j} B(j-\beta, D-j+1) = \frac{\Gamma(D+1)\Gamma(j-\beta)}{\Gamma(j+1)\Gamma(D+1-\beta)} \asymp D^\beta j^{-1-\beta}$$

gives the desired mass. On $[b, 1/2]$, the binomial factor decreases with p and, after multiplication by $\binom{D}{j}$, is at most $\exp[-D\,\mathrm{KL}(\rho\|b)]$. The remaining integrable weight makes this contribution exponentially small. This proves (5) and its stated extended range.

Sum (9) over the odd distances above r. Dropping the parity restriction and using the binomial Markov bound gives

$$\tau_r \le C D^{-\beta} \int_0^\infty u^{-1-\beta} \min\{1, Du/(r+1)\}\, du \le C(r+1)^{-\beta}.$$

For the lower bound, sum (5) over the odd j in $(r, 2r+2]$, using a fixed cutoff strictly below $D/2$ as needed at the endpoint. This yields $cr^{-\beta}$; bounded r are absorbed by a change of constants. Equation (7) follows from (5), (6), and (4).

For the second moment, use paired multiplicities to write $s_{2,r} = \sum_{j>r,\ j \text{ odd}} \binom{D}{j} a_j^2$. Up to an odd cutoff b_D near $D/4$, its terms are at most $Cj^{-2\beta-2}/\binom{D}{j}$. Consecutive odd inverse binomial coefficients have ratio

$$\frac{(j+1)(j+2)}{(D-j)(D-j-1)},$$

uniformly less than a fixed constant below one in this range. The sum is therefore bounded by a constant times its first term, at $j = r+2$. Beyond b_D, monotonicity gives the bound $a_{b_D}\tau_{b_D} \le CD^{-2\beta-1}/\binom{D}{b_D}$, which is exponentially smaller than the claimed bound uniformly for $r \le D/8$. This proves (8). □

Finally, two-step binomial ratios give, uniformly for odd $r \le D/8$,

$$\binom{D}{r} \le D_r \le C\binom{D}{r}, \qquad \frac{D_r}{\binom{D}{r+2}} \le C(r/D)^2. \tag{11}$$

All constants used here can be fixed independently of a subsequently chosen small positive limit $r/d \to \alpha$.

3 Random tail covariance matrices

Let Q be the covariance kernel retaining only degrees greater than an odd cutoff r, and write $\tau = \tau_r$, $\ell = \lambda_{r+1}$, and $s_2 = s_{2,r}$. This section proves the sample event needed later, without assuming that a Gaussian approximation remains valid after conditioning. The entire argument concerns the exact limiting Gaussian covariance.

Lemma 3.1 (Far-pair decay)**.** *There are constants $c_0, C > 0$ such that, when $r \le D/8$ is odd,*

$$\max_{d/4 \le \mathrm{dist}(x,y) \le 3d/4} |Q(x,y)| \le \rho_r := C\sqrt{D}\, e^{-r/8} + Ce^{-c_0 D}.$$

Proof. The full-cube Krawtchouk polynomial satisfies

$$\sum_{j=0}^{D} \mathsf{K}_j(h;D) z^j = (1+z)^{D-h}(1-z)^h.$$

If $\delta D \le h \le (1-\delta)D$, put $b = \min(h, D-h)$. On $|z| = t$ the generating function is bounded in absolute value by

$$(1+t^2)^b (1+t)^{D-2b} \le (1+t)^D \exp\left[-\frac{2\delta D t}{(1+t)^2}\right].$$

Cauchy's coefficient bound at $t = j/(D-j)$ and the uniform Stirling lower bound for $\binom{D}{j}$ therefore give, for $1 \le j < D$,

$$\frac{|\mathsf{K}_j(h;D)|}{\binom{D}{j}} \le C\sqrt{D}\exp[-2\delta j(1-j/D)]. \tag{12}$$

For reduced-cube points at distance in $[d/4, 3d/4]$, their augmented full-cube distance lies in $[D/8, 7D/8]$ for large d. Because r is odd, the exact tail is $Q(x,y) = \sum_{j>r,\ j \text{ odd}} a_j \mathsf{K}_j(h;D)$. For $r < j \le D - r$, (12) gives a bound $C\sqrt{D}e^{-r/8}$ after summation, since the total spectral mass is one. For $j > D - r$, monotonicity and an odd cutoff b_D near $D/4$ give

$$\sum_{j>D-r} \binom{D}{j} a_j \le a_{b_D} \sum_{k<r} \binom{D}{k} \le CD^{-\beta-1/2}\exp[-D(h(1/4) - h(1/8)) + O(1)].$$

The positive entropy gap absorbs the polynomial factor and proves the claim. Adjust constants for bounded d. □

Lemma 3.2 (A tail Gram deviation bound)**.** *Let $X_1, \dots, X_n$ be independent uniform cube points, with $1 \ll n \le M/4$. Let*

$$B_{\mathrm{near}} = \#\{y : \mathrm{dist}(x,y) \notin [d/4, 3d/4]\} \le 2e^{dh(1/4)}, \qquad q = 2\log M.$$

The count is independent of x. For every $z > 0$,

$$\begin{aligned}&\mathbb{P}\{\|Q_{SS} - \tau I_n\|_{\mathrm{op}} > z\tau\} \\ &\quad \le \frac{C}{z^2}\left[\frac{q\rho_r}{\tau} + \frac{\sqrt{nqs_2}}{\tau} + \frac{n\ell}{\tau} + \frac{n(1+B_{\mathrm{near}})}{M}\right]^2 + \frac{n^2(1+B_{\mathrm{near}})}{M}.\end{aligned} \tag{13}$$

Here the sampled matrix includes repeated training locations.

Proof. Let $J = Q - \tau I_M$, a zero-diagonal Hermitian matrix. Form a deterministic mask J_{far} retaining only pairs at distances in $[d/4, 3d/4]$. Since Q is positive semidefinite with constant diagonal τ, every entry has absolute value at most τ. Thus

$$\begin{aligned} \|J_{\mathrm{far}}\|_{\max} &\le \rho_r, \\ \|J_{\mathrm{far}}\|_{1\to 2}^2 &\le M s_2, \\ \|J_{\mathrm{far}}\|_{\mathrm{op}} &\le M\ell + \tau + \tau B_{\mathrm{near}}. \end{aligned}$$

For the second inequality, Parseval gives squared column norm $Ms_2 - \tau^2$ for J, and entrywise masking only decreases it. For the third, $\|J\| \le M\ell + \tau$, while the removed matrix has maximum absolute row sum at most τB_{near}. The entropy bound on B_{near} is the elementary binomial tail bound on distances to a fixed cube point.

Let R retain each of the M coordinates independently with probability $p = 2n/M$. The zero-diagonal case of Tropp's random principal-submatrix theorem [5, Theorem 1.1], at moment order two and $q = 2\log M$, states

$$(\mathbb{E}\|RJR\|^2)^{1/2} \le C\left[q(\mathbb{E}\|RJR\|_{\max}^2)^{1/2} + \sqrt{pq}(\mathbb{E}\|JR\|_{1\to 2}^2)^{1/2} + p\|J\|\right].$$

Apply it to J_{far} and substitute the preceding bounds:

$$(\mathbb{E}\|RJ_{\mathrm{far}}R\|^2)^{1/2} \le C[q\rho_r + \sqrt{nqs_2} + n\ell + (n/M)\tau(1 + B_{\mathrm{near}})]. \tag{14}$$

The logarithm here is that of the original dimension M, not the sample size.

To pass to a uniform subset of size n, use a uniform permutation of all cube points, independently of $B \sim \mathrm{Bin}(M, 2n/M)$. Its first B elements form the Bernoulli subset. On $B \ge n$, the first n principal submatrix is a compression of the larger one, so its operator norm cannot be larger. The size-n event is independent of B, and $\mathbb{P}(B \ge n) \ge 1/2$ for large n. Thus Markov's inequality applied to (14) gives the same fixed-size probability bound with a constant factor loss.

Conditional on no collisions, an independent size-n sample is a uniform size-n subset, up to an irrelevant ordering. Collisions have probability at most $n(n-1)/(2M)$. The probability of any sampled pair violating the distance condition is at most $n^2 B_{\mathrm{near}}/M$. On their complement, the sampled matrices J and J_{far} agree. Adding these exceptional probabilities proves (13). □

Corollary 3.3 (Simultaneous spectral and Gram control)**.** *Fix $A > 0$ and arbitrary constants $\eta, \xi > 0$. There is a sufficiently small $\alpha > 0$ such that, whenever r_d is odd with $r_d/d \to \alpha$ and $n_d = \lceil AD_{r_d}\log(D_{r_d}+1)\rceil$, for all sufficiently large d,*

$$\frac{n_d\lambda_{r_d+1}}{\tau_{r_d}} \le \xi, \qquad \mathbb{P}_S\{\|Q_{SS}\|_{\mathrm{op}} > 2\tau_{r_d}\} \le \eta.$$

Proof. Equations (7), (8), and (11) give

$$\begin{aligned} \frac{n_d\ell}{\tau} &\le C_L A\alpha^2\log(e/\alpha) + o(1), \\ \frac{n_d q s_2}{\tau^2} &\le C_L A\alpha\log(e/\alpha) + o(1). \end{aligned}$$

Also $q\rho_r/\tau$ tends to zero exponentially. Choose α small enough that $2h(\alpha) < \log 2 - h(1/4)$. Then all near-pair and collision terms in (13) tend to zero exponentially, because $n_d = \exp((h(\alpha)+o(1))d)$. The remaining bracket in (13) is at most

$$C_L\left[\sqrt{A\alpha\log(e/\alpha)} + A\alpha^2\log(e/\alpha)\right] + o(1).$$

It tends to zero as $\alpha \downarrow 0$ after A is fixed. Taking $z = 1$ in that bound and then choosing α still smaller proves both assertions. □

4 The actual Gaussian sign posterior

Let $\mathcal{X} = \{-1, 1\}^d$, $M = 2^d$, and $f(x) = \prod_{j=1}^d x_j$. All function norms and expectations on $\mathcal{X}$ use its uniform probability measure. Let G be a nondegenerate centered Gaussian vector indexed by $\mathcal{X}$, with covariance matrix K. The training inputs $X_1, \dots, X_n$ are independent and uniform. We condition on

$$\mathcal{A}_S = \{f(X_i)G(X_i) > 0 : 1 \le i \le n\}.$$

The prediction at x is its actual conditional correct-label probability $q_S(x) = \mathbb{P}(f(x)G(x) > 0 \mid \mathcal{A}_S)$, and we write

$$F_\theta(S) = \mathbb{P}_x\{q_S(x) \le \theta\}, \qquad \theta = \frac{13}{20}, \qquad c = \sqrt{2\pi}\left(\frac{1}{e} - \frac{7}{20}\right) > 0.$$

The threshold $\theta < 2/3$ supplies a strict buffer for passage from limiting Gaussian posteriors to finite-width posterior probabilities.

Suppose the Walsh characters are eigenfunctions of the normalized covariance operator, with degree-j eigenvalue λ_j. Write $V_r = \operatorname{span}\{\chi_U : |U| \le r\}$ and $D_r = \dim V_r$. For $r < d$, let Q be the covariance contributed by the complementary Fourier modes, and put

$$\tau = Q(x, x) = \sum_{j>r} \binom{d}{j} \lambda_j, \qquad \ell = \max_{j>r} \lambda_j.$$

The matrix eigenvalues of Q are $M\lambda_j$; this normalization is used throughout. Define $\|h\|_K^2 = h^T K^{-1} h$, and write $u_+ = \max(u, 0)$ and $u_- = \max(-u, 0)$.

Lemma 4.1 (Mean norm and a confidence margin)**.** *Let $m(x) = \mathbb{E}[G(x) \mid \mathcal{A}_S]$, and let $s \le n$ be the number of distinct sampled inputs. Then*

$$\|m\|_K^2 \le s, \qquad f(X_i)m(X_i) > 0.$$

Let $\sigma_x^2 = \operatorname{Var}(G(x) \mid G_S)$, where S here lists the distinct inputs. Whenever $\sigma_x > 0$ and $q_S(x) > \theta$,

$$f(x)m(x) > c\sigma_x.$$

Proof. Whiten the Gaussian training vector as $G_S = K_{SS}^{1/2}\xi$. The sign constraints form a homogeneous cone in ξ. Conditioning leaves the squared Gaussian radius with mean s, so Jensen's inequality and Gaussian regression give

$$m_S^T K_{SS}^{-1} m_S \le s, \qquad m = K_{\mathcal{X},S} K_{SS}^{-1} m_S.$$

The RKHS norm of this regression vector is the preceding quadratic form. Its observed signed coordinates are strictly positive.

For the confidence claim, the scalar conditional variable $Y = f(x)G(x)$ has a log-concave density: it is a marginal of a Gaussian restricted to a convex set, by the marginal log-concavity theorem [7]. If $\mu = \mathbb{E}Y$ and F_Y is its cdf, log-concavity and Jensen's inequality imply

$$\log F_Y(\mu) \ge \mathbb{E} \log F_Y(Y) = -1.$$

Consequently $\mathbb{P}(Y > \mu) \le 1 - e^{-1} < \theta$. Conditioning first on the full training vector shows that the density of Y is a mixture of Gaussian densities with common variance σ_x^2, and is therefore at most $(\sqrt{2\pi}\sigma_x)^{-1}$. If $q_S(x) > \theta$, then $\mu > 0$ and

$$\theta < \mathbb{P}(Y > 0) \le 1 - e^{-1} + \frac{\mu}{\sqrt{2\pi}\sigma_x}.$$

This gives the stated margin. □

Lemma 4.2 (Most residual variances remain large)**.** *For every training sample,*

$$\mathbb{P}_x\{\sigma_x^2 < \tau/2\} \le \frac{2n\ell}{\tau}.$$

Also, if p is the degree-r Fourier truncation of m and $t = m - p$, then $\|t\|_2^2 \le n\ell$.

Proof. Write $G = G_{\le r} + G_{>r}$ as a sum of independent fields. Conditioning additionally on the full low-degree field can only decrease conditional Gaussian variance. The remaining covariance is

$$Q - Q_{\mathcal{X},S} Q_{SS}^{\dagger} Q_{S,\mathcal{X}}.$$

Its subtracted term is positive semidefinite, bounded above by Q, and has rank at most $s \le n$. Its normalized trace is at most $n\ell$. Markov's inequality proves the first assertion; it includes observed points automatically. Fourier orthogonality and the mean norm bound prove $\|t\|_2^2 \le \ell\|m\|_K^2 \le n\ell$. □

5 Uniform sampling detects the negative part

Lemma 5.1 (Uniform ReLU sampling on the cube)**.** *For every fixed $\delta, \eta \in (0,1)$, there is a universal constant C such that, if*

$$n \ge C\eta^{-2} D_r \log\left(\frac{D_r}{\delta\eta}\right),$$

then with probability at least $1-\delta$, simultaneously for all $p \in V_r$,

$$\left|\frac{1}{n}\sum_{i=1}^{n}(-f(X_i)p(X_i))_+ - \mathbb{E}_x(-f(x)p(x))_+\right| \le \eta\mathbb{E}_x|p(x)|.$$

In particular, when $r < d$ and $\eta = 1/4$, the empirical negative part is at least $\mathbb{E}|p|/4$.

Proof. Let A have rows $f(x)(\chi_U(x))_{|U|\le r}$, one for each cube point. Then $A^TA = MI$ and every row has squared norm D_r. The ℓ_1 Lewis weights of A are exactly D_r/M: substitution in their defining equation

$$w_x^2 = A_x(A^T \operatorname{diag}(w^{-1})A)^{-1}A_x^T$$

verifies this constant solution, and Lewis weights are unique. The additive ReLU coreset theorem of Mai, Musco, and Rao [4, Theorem 5] therefore samples precisely the uniform distribution on the cube and gives the stated estimate. Its sample bound is $O(\eta^{-2}D_r\log(D_r/(\delta\eta)))$; no additional logarithmic factor is required. Finally $\mathbb{E}fp = 0$, because $r < d$, so $\mathbb{E}(-fp)_+ = \mathbb{E}|p|/2$. □

Proposition 5.2 (A deterministic reduction to a tail Gram bound)**.** *Suppose a design obeys the conclusion of Lemma 5.1 with $\eta = 1/4$, and also*

$$\|Q_{SS}\|_{\mathrm{op}} \le B\tau, \tag{15}$$

$$\frac{1}{n}\#\{i : f(X_i)p(X_i) < 0\} \le \mathbb{P}_x\{f(x)p(x) \le 0\} + u \quad \text{for all } p \in V_r. \tag{16}$$

Here $B \ge 1$, and the matrix in (15) lists the training sample with multiplicity. Put $\zeta = n\ell/\tau$. If

$$u \le \frac{c^2}{2048B}, \qquad \left(2 + \frac{8}{c^2}\right)\zeta \le \frac{c^2}{2048B},$$

then the actual Gaussian sign posterior satisfies

$$F_\theta(S) \ge \frac{c^2}{1024B}.$$

Proof. Let $m = p + t$ be the low/high decomposition of its conditional mean, and write $E_p = \mathbb{P}_x\{fp \le 0\}$. Training consistency implies

$$(f(X_i)p(X_i))_- \le |t(X_i)|\mathbf{1}_{\{f(X_i)p(X_i)<0\}}.$$

On the tail subspace $\|t\|_Q = \|t\|_K$. RKHS evaluation and $\|t\|_K^2 \le n$ give

$$\frac{1}{n}\sum_i t(X_i)^2 \le \frac{\|t\|_K^2}{n}\|Q_{SS}\|_{\mathrm{op}} \le B\tau.$$

Combining the ReLU estimate, Cauchy–Schwarz, and (16) yields

$$\mathbb{E}|p| \le 4\sqrt{B\tau(E_p + u)}. \tag{17}$$

Lemma 4.2 and Markov's inequality show that, outside sets of total measure at most $a_0 = (2+8/c^2)\zeta$, both $\sigma_x \ge \sqrt{\tau/2}$ and $|t(x)| \le c\sqrt{\tau}/(2\sqrt{2})$ hold. At every remaining input with confidence greater than θ, Lemma 4.1 gives

$$f(x)p(x) > \frac{c\sqrt{\tau}}{2\sqrt{2}}.$$

Writing $e = F_\theta(S) + a_0$, we obtain $E_p \le e$ and

$$\frac{\mathbb{E}|p|}{\sqrt{\tau}} \ge \frac{c}{2\sqrt{2}}(1 - e).$$

If $e \le 1/2$, comparison with (17) gives $e + u \ge c^2/(512B)$, and hence $F_\theta(S) \ge c^2/(1024B)$ under the stated assumptions. If $e > 1/2$, the same claimed lower bound follows directly from the upper bound on a_0. □

6 An exponentially large sample schedule

The preceding proposition is deterministic. The remaining ingredients are a standard VC estimate and the spectral/random-Gram estimates proved in the preceding sections. We state their exact required form to make the scope of the conclusion explicit.

Theorem 6.1 (Failure along an exponential sample schedule)**.** *Fix the depth $L \ge 2$. For the covariance*

$$K(x, y) = H^{\circ(L-1)}\left(\frac{1 + x \cdot y}{d + 1}\right), \qquad H(t) = \frac{2}{\pi}\arcsin t,$$

there are constants $A > 0$, $\alpha \in (0, 1/2)$, and $\varepsilon > 0$ such that the following holds. Let r_d be an odd integer with $r_d/d \to \alpha$, and set

$$n_d = \lceil AD_{r_d}\log(D_{r_d} + 1)\rceil.$$

For all sufficiently large d, the Gaussian sign posterior trained on n_d independent uniform parity examples satisfies

$$\mathbb{P}_S\{F_{13/20}(S) \ge \varepsilon\} \ge \frac{3}{4}.$$

In particular $n_d = \exp((h(\alpha) + o(1))d)$, where $h(\alpha) = -\alpha\log\alpha - (1-\alpha)\log(1-\alpha) > 0$. The same conclusion holds with $F_{13/20}(S)$ replaced by $\liminf_{N\to\infty} F_N(S)$ in any finite-width model whose conditional prediction probabilities converge to this Gaussian sign posterior. This includes the negative-γ scaling width limit.

Proof. Fix $B = 2$ and $\varepsilon = c^2/(1024B)$. Set $u = c^2/(2048B)$ and $\delta = 1/20$. First choose A large enough for Lemma 5.1 with $\eta = 1/4$ and this δ. The class $\{x : f(x)p(x) < 0,\ p \in V_r\}$ has VC dimension at most $D_r + 1$. Thus the uniform VC theorem [6] gives (16) with probability at least $1 - \delta$ for all large d, since $n_d/D_{r_d} \to \infty$.

The spectral estimates give, for odd r and r/d sufficiently small,

$$\frac{n_d \lambda_{r+1}}{\tau_r} \le C_L A \alpha^2 \log(1/\alpha) + o(1).$$

Corollary 3.3 gives $\mathbb{P}_S\{\|Q_{SS}\|_{\mathrm{op}} > 2\tau_r\} \le \delta$ for all large d after α is made sufficiently small, depending on A, L, δ. Decrease α further so that the preceding spectral ratio also meets Proposition 5.2. A union bound over the three sampling events proves a probability of at least $1 - 3\delta > 3/4$.

For the final assertion, fix d and the sampled training set. At every input with limiting correct-label probability at most $13/20$, the finite-width correct-label probability is eventually below $2/3$. The cube is finite, so this holds simultaneously at every such input, which proves $\liminf_N F_N(S) \ge F_{13/20}(S)$. □

The theorem concerns the displayed sample schedule. It does not assert failure at every smaller sample size, and no monotonicity of posterior confidence in sample size is assumed. The later paper [1] covers all intervening sample sizes by an additional middle-band argument. The proof also uses convexity of Gaussian sign conditioning; it does not establish the analogous exponential conclusion at $\gamma = 0$ or $0 < \gamma < 1$.

References

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